Lecture 13. Refinements of the Proportional Hazards Model

Time-Dependent Covariates, Piecewise and Stratified PH Models

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1 Time-Dependent Covariates

Motivation. Covariates such as blood pressure or a post-transplant event change during follow-up; they also give a way to test, and to relax, proportional hazards.

Types: discrete (e.g. 0 until an event, 1 thereafter) and continuous (blood pressure, cholesterol, body mass index, tumour size or its rate of change).

Definition 13.1: PH model with time-dependent (TD) covariates

For $\mathbf{Z}(t) = [Z_1(t), \ldots, Z_p(t)]'$,

$$h(x \mid \mathbf{Z}(t), t \le x) = h_0(x) \exp\{\boldsymbol{\beta}'\mathbf{Z}(x)\},$$

conditioning on the covariate history up to x, with the hazard depending only on the current value $\mathbf{Z}(x)$. $\mathbf{Z}(t)$ is **predictable**: known just before t.

1.1 Estimation

Data: n i.i.d. $\{T_j, \delta_j, [\mathbf{Z}_j(t), 0 \le t \le T_j]\}$, with $\mathbf{Z}_j(t)$ known whenever subject j is under observation. With $\mathbf{Z}_{(i)}(t)$ the covariate of the subject failing at t_i,

$$PL(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp\{\boldsymbol{\beta}'\mathbf{Z}_{(i)}(t_i)\}}{\sum_{j \in R(t_i)} \exp\{\boldsymbol{\beta}'\mathbf{Z}_j(t_i)\}}. \tag{9.2.1}$$

Each risk-set member enters with its covariate value *at* t_i. $\hat{\boldsymbol{\beta}} = \arg\max PL$ is consistent and asymptotically normal with $\widehat{V}(\hat{\boldsymbol{\beta}}) = I^{-1}(\hat{\boldsymbol{\beta}})$; all inference of Lectures 11–12 (local tests, tie corrections, estimating H_0 and $S(x \mid \mathbf{Z}(t), t \le x)$) carries over.

Example 13.2: bone marrow transplant, intermediate events

Disease-free survival in three groups (ALL, AML low-risk, AML high-risk; Lecture 9). Three intermediate events of recovery are coded as TD indicators:

$Z_A(t) = I(t \geq$ time of acute graft-versus-host disease), $Z_C(t) = I(t \geq$ time of chronic graft-versus-host dise

To avoid bias in the group comparison, each is tested in a model already holding the group indicators $Z_1 = I(\text{AML low})$, $Z_2 = I(\text{AML high})$; Wald tests of the TD coefficient:

Model	b	SE(b)	Wald χ^2	p
Z_1	−0.5516	0.2880	3.669	0.0554
Z_2	0.4338	0.2722	2.540	0.1110
$Z_A(t)$	0.3184	0.2851	1.247	0.2642
Z_1	−0.6225	0.2962	4.4163	0.0356
Z_2	0.3657	0.2685	1.8548	0.1732
$Z_C(t)$	−0.1948	0.2876	0.4588	0.4982
Z_1	−0.4962	0.2892	2.9435	0.0862
Z_2	0.3813	0.2676	2.0306	0.1542
$Z_P(t)$	−1.1297	0.3280	11.8657	0.0006

Only platelet recovery matters: after it, the rate of relapse or death is $e^{-1.1297} = 0.323$ times the rate before.

1.2 External versus internal covariates

- **External** (not affected by the failure process): air pollution, temperature, public-health policy.
- **Internal** (measured on the subject, related to failure): tumour size, blood biomarkers, cholesterol, ECG.

An internal $\mathbf{Z}(t)$ exists only while $T \geq t$, so

$$S(x \mid \mathbf{Z}(t), t \leq x) = \exp\Big\{ - \int_0^x h_0(t) e^{\boldsymbol{\beta}' \mathbf{Z}(t)} \, \mathrm{d}t \Big\}$$

is not well defined for all $0 \leq x \leq \infty$; with external covariates it is. $\hat{\boldsymbol{\beta}}$ is always valid as a log hazard ratio, but $\hat{S}(x \mid \mathbf{Z}(t), t \leq x)$ may be meaningless for internal covariates (the usual case in biomedicine).

Uses of TD covariates: covariates measured over time; testing PH for fixed covariates; piecewise PH models.

2 Checking Proportional Hazards

To test PH for a fixed covariate Z_1 (i.e. $h(t \mid Z_1 = z_1)/h(t \mid Z_1 = z_1^*)$ free of t), create $Z_2(t) = Z_1 \, g(t)$ for a known g (e.g. $\log t$, t) and fit

$$h(t \mid Z_1, Z_2(t)) = h_0(t) \exp\{\beta_1 Z_1 + \beta_2 Z_2(t)\}, \qquad \frac{h(t \mid Z_1 = z_1)}{h(t \mid Z_1 = z_1^*)} = \exp\{\beta_1 (z_1 - z_1^*) + \beta_2 g(t)(z_1 - z_1^*)\}.$$

PH holds iff $\beta_2 = 0$: a local test.

Example 13.3: catheter placement (Lecture 12 data)

$Z_1 = I(\text{percutaneous})$, event = first exit-site infection.

- **Graphical**: under PH, $H(t \mid Z=1) = e^{\beta} H(t \mid Z=0)$, so $\ln \tilde{H}(t \mid Z=1) - \ln \tilde{H}(t \mid Z=0)$ (Nelson–Aalen) should be roughly constant at β. The plot drifts downward: strong suggestion of non-proportionality.
- **Formal**: $Z_2(t) = Z_1 \ln t$ gives relative risk $\exp(\beta_1 + \beta_2 \ln t) = t^{\beta_2} e^{\beta_1}$, constant only if $\beta_2 = 0$. LR $= 12.22$ ($p = 0.0005$), Wald $= 6.19$ ($p = 0.013$): the hazards are not proportional.

3 Piecewise PH Model

Definition 13.4: Change point τ

With $Z_2(t) = Z_1\, I(t > \tau)$,

$$h(t \mid \mathbf{Z}(t)) = h_0(t) e^{\beta_1 Z_1 + \beta_2 Z_2(t)} = \begin{cases} h_0(t) e^{\beta_1 Z_1}, & t \le \tau, \\ h_0(t) e^{(\beta_1+\beta_2) Z_1}, & t > \tau, \end{cases}$$

so e^{β_1} ($e^{\beta_1+\beta_2}$) is the relative risk of $Z_1 = 1$ versus $Z_1 = 0$ before (after) τ. Equivalently, with $Z_2(t) = Z_1 I(t > \tau)$ and $Z_3(t) = Z_1 I(t \le \tau)$, $h = h_0(t) e^{\theta_2 Z_2(t) + \theta_3 Z_3(t)}$ has the same likelihood with $\theta_3 = \beta_1$, $\theta_2 = \beta_1 + \beta_2$.

Choice of τ: evaluate the log PL at the distinct failure times and take the maximizer. PH can then be re-tested before and after τ, and the procedure repeated if it fails.

Example 13.5: catheter placement, change point

Log partial likelihood by candidate τ:

τ	log PL	τ	log PL	τ	log PL
0.5	−97.878	5.5	−100.493	11.5	−102.168
1.5	−100.224	6.5	−98.856	15.5	−100.829
2.5	−97.630	8.5	−100.428	16.5	−101.477
3.5	**−97.500**	9.5	−101.084	18.5	−102.059
4.5	−99.683	10.5	−101.668	23.5	−102.620

$\tau = 3.5$ is optimal, and

$$h(t \mid Z_1) = \begin{cases} h_0(t) \exp(1.08 Z_1), & t \le 3.5 \quad (p = 0.17), \\ h_0(t) \exp(-2.09 Z_1), & t > 3.5 \quad (p = 0.006). \end{cases}$$

No evidence of a difference up to 3.5 months; afterwards percutaneous catheters are less likely to become infected.

Remark 13.6: Pros and cons

Separate Cox fits on different periods can be very informative, but the estimates are more variable, power drops, multiplicity grows; less attractive with small or medium samples or heavy censoring.

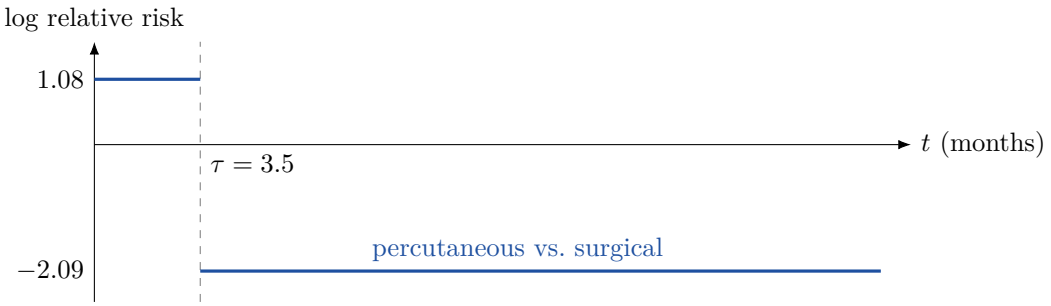


Figure 1: Fitted piecewise PH model for catheter placement: the log hazard ratio jumps at τ.

4 Stratified PH Model

Problem: PH fails for a covariate Z_0. **Solution**: if Z_0 is not of key interest, stratify on it and keep PH for the other covariates $\mathbf{Z}(t)$.

Definition 13.7: Stratified Cox model

If Z_0 takes s values, split the data into s strata and assume

$$h_j(t \mid \mathbf{Z}(t)) = h_{0j}(t)\exp\{\boldsymbol{\beta}'\mathbf{Z}(t)\}, \qquad j = 1, \ldots, s, \tag{9.3.1}$$

with stratum-specific baselines and common $\boldsymbol{\beta}$. The log PL is

$$LL(\boldsymbol{\beta}) = LL_1(\boldsymbol{\beta}) + LL_2(\boldsymbol{\beta}) + \cdots + LL_s(\boldsymbol{\beta}), \tag{9.3.2}$$

LL_j built from stratum j alone (risk sets within strata).

Theorem 13.8: Testing a common β

Under the alternative $h_j(t \mid \mathbf{Z}(t)) = h_{0j}(t)\exp\{\boldsymbol{\beta}_j'\mathbf{Z}(t)\}$, test $H_0 : \boldsymbol{\beta}_1 = \cdots = \boldsymbol{\beta}_s$. With $\hat{\boldsymbol{\beta}}_j = \arg\max LL_j(\boldsymbol{\beta}_j)$ and $\hat{\boldsymbol{\beta}} = \arg\max LL(\boldsymbol{\beta})$ in (9.3.2),

$$\chi^2_{LR} = 2\Big[\sum_{j=1}^{s} LL_j(\hat{\boldsymbol{\beta}}_j) - \sum_{j=1}^{s} LL_j(\hat{\boldsymbol{\beta}})\Big] \;\approx\; \chi^2_{(s-1)p} \quad \text{under } H_0.$$

Coefficient by coefficient ($s = 2$), $H_0 : \beta_{1i} = \beta_{2i}$:

$$\chi^2_{W,i} = \frac{(\hat{\beta}_{1i} - \hat{\beta}_{2i})^2}{\mathrm{SE}^2(\hat{\beta}_{1i}) + \mathrm{SE}^2(\hat{\beta}_{2i})} \;\approx\; \chi^2_1.$$

Example 13.9: bone marrow transplant, stratified on MTX

Covariates: $Z_P(t)$; group dummies g_2, g_3; patient and donor age centred at 28 and their interaction; FAB status. Earlier (Lecture 10) the groups were compared by a stratified log-rank test controlling for MTX; the Cox model now controls for all of these under PH, stratified on MTX ($p = 7$, $s = 2$). Separate fits give log PL -80.45 (MTX) and -219.64 (no MTX); the stratified fit gives -303.5:

$$\chi^2_{LR} = 2(-80.45 - 219.64 + 303.5) = 6.82 < \chi^2_{7;0.95} = 14.07$$

(the slide prints 6.12; either way) H_0 is not rejected.

Matched pairs: treat each pair as a stratum. For the 21 6-MP/placebo pairs (followed to relapse or end of study), all tests show a significant difference in relapse rates.

Remark 13.10: Handling non-proportionality

Piecewise PH (separate time periods); stratification on the offending factor; time-dependent effects $\beta(t)$; a different model class.

5 Selected exercises

Motivation. What a time-varying coefficient $\beta_1 + \beta_2 g(t)$ means, and why a matched-pair stratified Cox fit reproduces the paired sign test.

Exercise 13.1 (Reading a $Z_1 g(t)$ fit)

In Example 13.3 with $g(t) = \ln t$, suppose $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$.
(a) Find the time t^* at which the fitted relative risk equals 1, and describe it before and after t^*.
(b) Show that the piecewise model of Definition 13.4 is the special case $g(t) = I(t > \tau)$.

Solution [Supplement].

(a) $t^{\hat{\beta}_2} e^{\hat{\beta}_1} = 1$ iff $\hat{\beta}_1 + \hat{\beta}_2 \ln t = 0$, i.e. $t^* = \exp(-\hat{\beta}_1/\hat{\beta}_2)$. As $\hat{\beta}_2 < 0$, the log relative risk decreases in t: above 1 before t^*, below 1 after; the same qualitative picture as the change-point fit.

(b) With $g(t) = I(t > \tau)$, $Z_2(t) = Z_1 g(t)$ and the log relative risk is β_1 for $t \le \tau$, $\beta_1 + \beta_2$ for $t > \tau$. ■

Exercise 13.2 (Matched pairs: stratified Cox score test)

Each pair $i = 1, \ldots, M$ is a stratum with members $Z = 1$ (treated) and $Z = 0$ (control), no ties. Show that the stratified Cox score test of $H_0 : \beta = 0$ is $(D_1 - D_0)^2/(D_1 + D_0)$, where D_1 (D_0) counts pairs in which the treated (control) member has an observed event first: the paired test of Lecture 10.

Solution [Supplement]. In a pair only the first time can be an event with both at risk; a later event has a risk set of one and contributes nothing. An informative pair contributes to LL_i the term $\beta Z_{(1)} - \log(1 + e^{\beta})$, so at $\beta = 0$

$$U_i(0) = Z_{(1)} - \tfrac{1}{2} = \pm\tfrac{1}{2}, \qquad I_i(0) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}.$$

Summing over the $D_1 + D_0$ informative pairs, $U(0) = \frac{1}{2}(D_1 - D_0)$, $I(0) = \frac{1}{4}(D_1 + D_0)$, and $U(0)^2/I(0) = (D_1 - D_0)^2/(D_1 + D_0)$. ■